

161. The area of the region bounded by the curves $x(1 + y^2) = 1$ and $y^2 = 2x$ is :

वक्रों $x(1 + y^2) = 1$ और $y^2 = 2x$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है :

- (A) $2\left(\frac{\pi}{2} - \frac{1}{3}\right)$ (B) $\frac{1}{2}\left(\frac{\pi}{2} - \frac{1}{3}\right)$ (C) $\frac{\pi}{4} - \frac{1}{3}$ (D) $\frac{\pi}{2} - \frac{1}{3}$

Ans. D

Sol. $C_1 : (1 + y^2) = \left(\frac{1}{x}\right)$

$$y^2 = \left(\frac{1}{x} - 1\right)$$

$$C_2 : y^2 = 2x$$

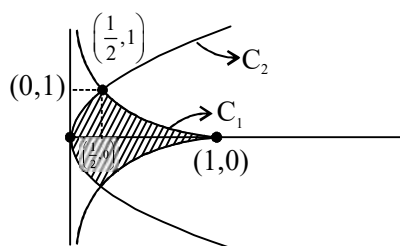
$$\left(\frac{1-x}{x}\right) = 2x$$

$$2x^2 = 1 - x$$

$$2x^2 + x - 1 = 0$$

$$(2x - 1)(x + 1) = 0$$

$$x = \frac{1}{2}, y = 1$$



$$\text{Resultant area with respect to y-axis} = 2 \int_0^1 \left[\left(\frac{1}{1+y^2} \right) - \left(\frac{y^2}{2} \right) \right] dy$$

$$\Rightarrow 2 \left[\tan^{-1}(y) - \frac{y^3}{6} \right]_0^1$$

$$\Rightarrow 2 \left[\frac{\pi}{4} - \frac{1}{6} \right]$$

$$\Rightarrow \left(\frac{\pi}{2} - \frac{1}{3} \right)$$

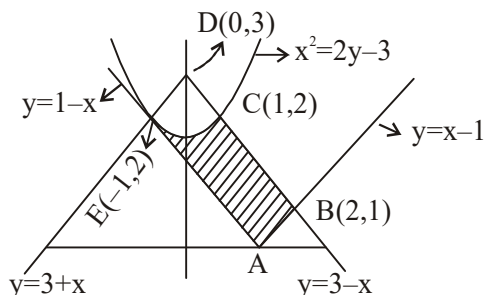
162. Let the area of the region $\{(x, y) : 2y \leq x^2 + 3, y + |x| \leq 3, y \geq |x - 1|\}$ be A. Then 6A is equal to :

माना कि क्षेत्र $\{(x, y) : 2y \leq x^2 + 3, y + |x| \leq 3, y \geq |x - 1|\}$ का क्षेत्रफल A है। तब 6A बराबर है :

- (A) 16 (B) 18 (C) 12 (D) 14

Ans. D

Sol.



$A = \text{area of rectangle } ABDE - \text{area of region } EDC$

$$A = 4 - 2 \int_0^1 (3 - x) - \left(\frac{x^2 + 3}{2} \right) dx$$

$$A = 4 - 2 \left(3 - \frac{1}{2} - \frac{1}{6} - \frac{3}{2} \right) = \frac{7}{3}$$

$$6A = 14$$

163. Let the area of enclosed between the curves $|y| = 1 - x^2$ and $x^2 + y^2 = 1$ be α . If $9\alpha = \beta\pi + \gamma$; β, γ are integers, then the value of $|\beta - \gamma|$ equals :

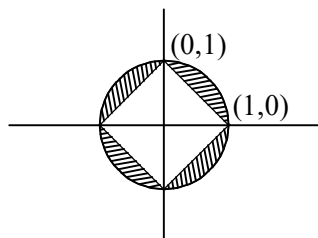
माना कि वक्रों $|y| = 1 - x^2$ और $x^2 + y^2 = 1$ के बीच घिरे क्षेत्र का क्षेत्रफल α है। यदि $9\alpha = \beta\pi + \gamma$; β, γ पूर्णांक है, तो $|\beta - \gamma|$ का मान बराबर है :

- (A) 18 (B) 33 (C) 27 (D) 15

Ans. B

Sol. $C_1 : |y| = 1 - x^2$

$$C_2 : x^2 + y^2 = 1$$



$\therefore$ Required Area

$$= \alpha = 4 \left[\text{Area of circle in 1st quad.} - \int_0^1 (1-x^2) dx \right]$$

$$= 4 \left[\frac{\pi}{4} - \left[x - \frac{x^3}{3} \right]_0^1 \right]$$

$$\alpha = \pi - \frac{8}{3}$$

$$\therefore 3\alpha = 3\pi - 8$$

$$\therefore 9\alpha = 9\pi - 24$$

$$\therefore \beta = 9, \gamma = -24$$

$$\therefore |\beta - \gamma| = 33$$

164. One of the points of intersection of the curves $y = 1 + 3x - 2x^2$ and $y = \frac{1}{x}$ is $\left(\frac{1}{2}, 2\right)$. Let the area of the region enclosed by the curves be $\frac{1}{24}(\ell\sqrt{5} + m) - n \log_e(1 + \sqrt{5})$, where $\ell, m, n \in \mathbb{N}$. Then $\ell + m + n$ is equal to :

वक्रों $y = 1 + 3x - 2x^2$ तथा $y = \frac{1}{x}$ का एक प्रतिच्छेदन $\left(\frac{1}{2}, 2\right)$ है। माना इन वक्रों से घिरे का क्षेत्रफल

$\frac{1}{24}(\ell\sqrt{5} + m) - n \log_e(1 + \sqrt{5})$ है, जहाँ $\ell, m, n \in \mathbb{N}$ है, तो $\ell + m + n$ का मान होगा :

(A) 32

(B) 29

(C) 31

(D) 30

Ans. D

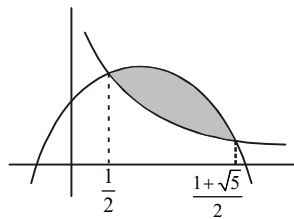
Sol. $2x^3 - 3x^2 - x + 1 = 0$

$$\Rightarrow (2x - 1)(x^2 - x - 1) = 0$$

$$\Rightarrow x = \frac{1}{2}, x = \frac{1 + \sqrt{5}}{2}$$

$$\text{Area} = \int_{\frac{1}{2}}^{\frac{1+\sqrt{5}}{2}} \left(1 + 3x - 2x^2 - \frac{1}{x} \right) dx$$

$$= \left[x + \frac{3x^2}{2} - \frac{2x^3}{3} - \ell \ln x \right]$$



$$\begin{aligned}
 &= \frac{\sqrt{5}+1}{2} + \frac{3}{8}(\sqrt{5}+1)^2 - \frac{1}{12}(\sqrt{5}+1)^3 - \ln\left(\frac{\sqrt{5}+1}{2}\right) - \left(\frac{1}{2} + \frac{3}{8} - \frac{1}{12} - \ln\frac{1}{2}\right) \\
 &= \frac{1}{24} \left[12(\sqrt{5}+1) + 9(\sqrt{5}+1)^2 - 2(\sqrt{5}+1)^3 - 12 - 9 + 2 \right] - \ln\left(\frac{\sqrt{5}+1}{2} \times 2\right) \\
 &= \frac{1}{24} \left[12(\sqrt{5}+1) + 9(6+2\sqrt{5}) - 2(5\sqrt{5}+1+3\sqrt{5}(\sqrt{5}+1)-19) \right] - \ln(\sqrt{5}+1) \\
 &= \frac{1}{24} [14\sqrt{5}+15] - \ln(\sqrt{5}+1) \\
 \therefore l &= 14, m = 15, n = 1 \\
 l + m + n &= 30
 \end{aligned}$$

165. The area enclosed between the curves $y = x|x|$ and $y = x - |x|$ is :

चक्रों $y = x|x|$ तथा $y = x - |x|$ के बीच घिरे क्षेत्र का क्षेत्रफल है :

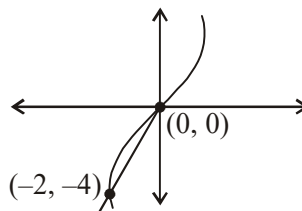
- (A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) 1

Ans. B

Sol. $y = x|x|$ & $y = x - |x|$

$$y = x|x| = \begin{cases} x^2, & x > 0 \\ -x^2, & x < 0 \end{cases}$$

$$y = x - |x| = \begin{cases} 0, & x > 0 \\ 2x, & x < 0 \end{cases}$$



point of intersections are (0, 0) & (-2, -4).

$$\therefore \text{Area} = \int_{-2}^0 (-x^2 - 2x) dx$$

$$= \left[\frac{-x^3}{3} - \frac{2x^2}{2} \right]_{-2}^0$$

$$= -\left[\frac{8}{3} - 4 \right] = \frac{4}{3} \text{ sq. unit}$$

166. Let the area of the region enclosed by the curves $y = 3x$, $2y = 27 - 3x$ and $y = 3x - x\sqrt{x}$ be A. Then 10A is equal to :

माना वक्रों $y = 3x$, $2y = 27 - 3x$ और $y = 3x - x\sqrt{x}$ से घिरे क्षेत्र का क्षेत्रफल A है, तो $10A$ बराबर है :

(A) 184

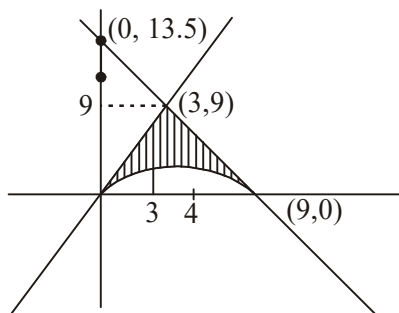
(B) 172

(C) 154

(D) 162

Ans. D

Sol.



$$y = 3x - x^{\frac{3}{2}}$$

$$y' = 3 - \frac{3}{2}\sqrt{x}$$

$$y' = 3 \left(1 - \frac{\sqrt{x}}{2} \right)$$

$$f'(x) \text{ ——— } + \text{ ——— } -$$

Area of shaded region

$$= \int_0^3 \left(3x - \left(3x - x^{\frac{3}{2}} \right) \right) dx + \int_3^9 \frac{27 - 3x}{2} - \left(3x - x^{\frac{3}{2}} \right) dx$$

$$= \int_0^3 x^{\frac{3}{2}} dx + \frac{1}{2} \int_3^9 (27 - 9x + 2x^{\frac{3}{2}}) dx = \left(\frac{2}{5} x^{\frac{5}{2}} \right)_0^3 + \frac{1}{2} \left[27x - \frac{9x^2}{2} + 2 \times \frac{2}{5} x^{\frac{5}{2}} \right]_3^9$$

$$= \frac{2}{5} \times 9\sqrt{3} + \frac{1}{2} \left[243 - \frac{729}{2} + \frac{4}{5} \times 81 \times 3 - 81 + \frac{81}{2} - \frac{4}{5} \times 9\sqrt{3} \right]$$

$$= \frac{1}{2} \left[\frac{486 - 729 - 81}{2} + \frac{972}{5} \right] = \frac{81}{5}$$

$$\Rightarrow 10A = 162$$

167. If the area of the region $\left\{ (x, y) : \frac{a}{x^2} \leq y \leq \frac{1}{x}, 1 \leq x \leq 2, 0 < a < 1 \right\}$ is $(\log_e 2) - \frac{1}{7}$ then the value of $7a - 3$ is equal to :

यदि क्षेत्र $\left\{ (x, y) : \frac{a}{x^2} \leq y \leq \frac{1}{x}, 1 \leq x \leq 2, 0 < a < 1 \right\}$ का क्षेत्रफल $(\log_e 2) - \frac{1}{7}$ है, तो $7a - 3$ का मान बराबर है :

(A) 1

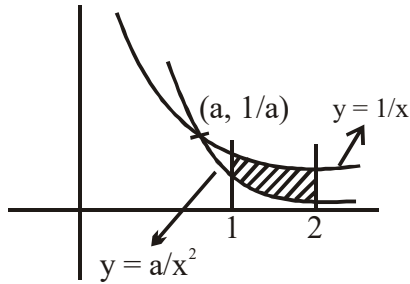
(B) 2

(C) 0

(D) -1

Ans. D

Sol. Required Area $= \int_1^2 \left(\frac{1}{x} - \frac{a}{x^2} \right) dx = \left(\ln x + \frac{a}{x^2} \right) = \ln 2 + \frac{a}{2} - a$



$$\text{Now, } \ln 2 + \frac{a}{2} - a = \ln 2 - \frac{1}{7}$$

$$\frac{a}{2} = \frac{1}{7} \Rightarrow 7a = 2 \text{ and } 7a - 3 = -1$$

168. The parabola $y^2 = 4x$ divides the area of the circle $x^2 + y^2 = 5$ in two parts. The area of the smaller part is equal to:

परवलय $y^2 = 4x$, वृत्त $x^2 + y^2 = 5$ के क्षेत्रफल को दो भागों में बाँटता है। छोटे भाग का क्षेत्रफल है :

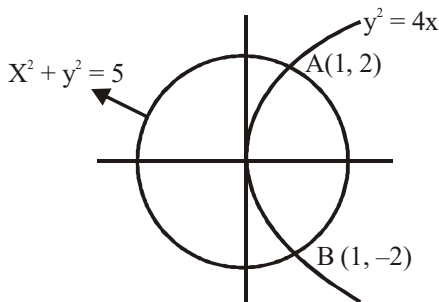
(A) $\frac{1}{3} + \sqrt{5} \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$

(B) $\frac{1}{3} + 5 \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$

(C) $\frac{2}{3} + 5 \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$

(D) $\frac{2}{3} + \sqrt{5} \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$

Ans. C



Sol.

$$\text{Area of smaller region} = \int_{-2}^2 \left(\sqrt{5-y^2} - \frac{y^2}{4} \right) dy$$

$$= 2 \int_0^2 \left(\sqrt{5-y^2} - \frac{y^2}{4} \right) dy$$

$$= 2 \left[\left(\frac{y}{2} \sqrt{5-y^2} + \frac{5}{2} \sin^{-1} \left(\frac{y}{\sqrt{5}} \right) \right) - \frac{1}{12} (y^3) \right]_0^2$$

$$= 2 \left[1 + \frac{5}{2} \sin^{-1} \left(\frac{2}{\sqrt{5}} \right) - \frac{2}{3} \right] = \frac{2}{3} + 5 \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$$

169. The area of the region $\left\{ (x, y) : y^2 \leq 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \neq 3 \right\}$ is :

क्षेत्र $\left\{ (x, y) : y^2 \leq 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \neq 3 \right\}$ का क्षेत्रफल है :

(A) $\frac{32}{3}$

(B) $\frac{16}{3}$

(C) $\frac{64}{3}$

(D) $\frac{8}{3}$

Ans. A

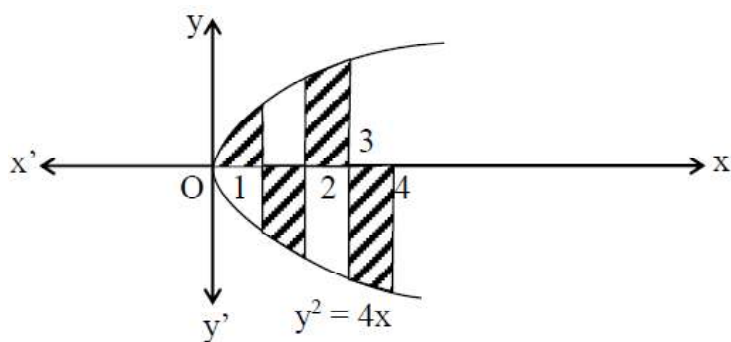
Sol. $\left\{ (x, y) : y^2 \leq 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \neq 3 \right\}$

$$y^2 \leq 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0$$

$$\text{Case (i) } y > 0 \Rightarrow \frac{x(x-1)(x-2)}{(x-3)(x-4)} < 0$$

$$x \in (1, 2) \cup (3, 4)$$

$$\text{Area} = \int_0^1 2\sqrt{x} dx + \left| \int_1^2 2\sqrt{x} dx \right| + \int_2^3 2\sqrt{x} dx + \left| \int_3^4 2\sqrt{x} dx \right|$$



$$= \frac{4}{3} \left(\left[\frac{3}{x^2} \right]_0^1 + \left[\frac{3}{x^2} \right]_1^2 + \left[\frac{3}{x^2} \right]_2^3 + \left[\frac{3}{x^2} \right]_3^4 \right)$$

$$A = \frac{4}{3} (1 + 2\sqrt{2} - 1 + 3\sqrt{3} - 2\sqrt{2} + 4\sqrt{4} - 3\sqrt{3}) = \frac{4}{3} \times 8 = \frac{32}{3}$$

170. The area enclosed by the curves $xy + 4y = 16$ and $x + y = 6$ is equal to :

वक्रों $xy + 4y = 16$ तथा $x + y = 6$ से घिरे क्षेत्र का क्षेत्रफल है :

- (A) $28 - 30 \log_e 2$ (B) $32 - 30 \log_e 2$ (C) $30 - 28 \log_e 2$ (D) $30 - 32 \log_e 2$

Ans. D

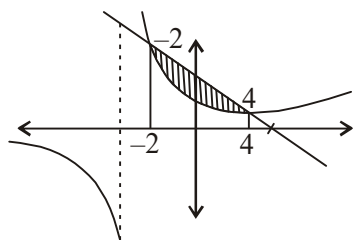
Sol. $y(x + 4) = 16$ & $y = 6 - x$

solve both curves,

$$(6 - x)(x + 4) = 16$$

$$6x - x^2 + 24 - 4x = 16$$

$$x^2 - 2x - 8 = 0 \Rightarrow x = -2, 4$$



$$\int_{-2}^4 \left(6 - x - \left(\frac{16}{x + 4} \right) \right) dx$$

$$\Rightarrow \left[6x - \frac{x^2}{2} - 16 \ln|x+4| \right]_{-2}^4$$

$$\Rightarrow (24 - 8 - 16 \ln 8) - (-12 - 2 - 16 \ln 2)$$

$$\Rightarrow 30 - 32 \ln 2$$

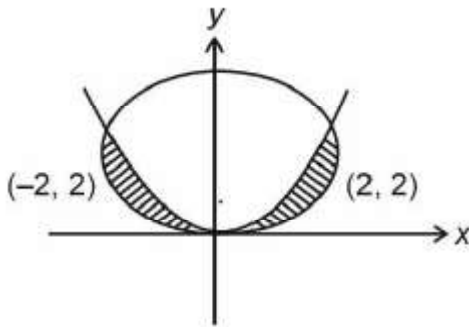
171. Area of the region $\{(x, y) : x^2 + (y - 2)^2 \leq 4, x^2 \geq 2y\}$ is :

क्षेत्र $\{(x, y) : x^2 + (y - 2)^2 \leq 4, x^2 \geq 2y\}$ का क्षेत्रफल :

- (A) $2\pi - \frac{16}{3}$ (B) $\pi + \frac{8}{3}$ (C) $\pi - \frac{8}{3}$ (D) $2\pi + \frac{16}{3}$

Ans. A

Sol. $x^2 + (y - 2)^2 \leq 4, x^2 \geq 2y$



$$\text{Area of required region} = 2 \left[\frac{1}{4} \pi (4) - \int_0^2 \sqrt{2} \cdot \sqrt{y} dy \right]$$

$$\Rightarrow 2 \left[\pi - \frac{\sqrt{2} \cdot y^{3/2}}{3/2} \right]_0^2$$

$$\Rightarrow 2 \left[\pi - \frac{2\sqrt{2}}{3} \cdot 2\sqrt{2} \right] = 2 \left[\pi - \frac{8}{3} \right] = 2\pi - \frac{16}{3}$$

172. The area of the region enclosed by the curve $y = x^3$ and its tangent at the point $(-1, -1)$ is :

वक्र $y = x^3$ तथा इसके बिंदु $(-1, -1)$ पर स्पर्श रेखा से घिरे क्षेत्र का क्षेत्रफल है :

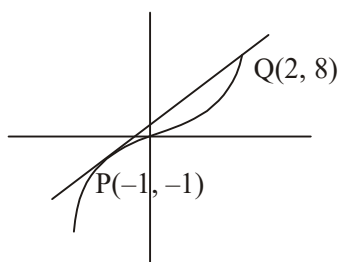
- (A) $\frac{23}{4}$ (B) $\frac{19}{4}$ (C) $\frac{27}{4}$ (D) $\frac{31}{4}$

Ans. C

Sol. Given $y = x^3$ _____(1)

$$\Rightarrow \frac{dy}{dx} = 3x^2$$

$$\left(\frac{dy}{dx} \right)_{(-1,-1)} = 3$$



Equation of tangent at $(-1, -1)$

$$(y + 1) = 3(x + 1)$$

$$y = 3x + 2$$
 _____(2)

Solving (1) and (2)

$$x^3 = 3x + 2$$

$$x^3 - 3x - 2 = 0$$

$\begin{array}{l} \nearrow -1 \\ \hline \nearrow -1 \\ \searrow \alpha \end{array}$

$-2 + \alpha = 0$
 $\alpha = 2$

$$Q \equiv (2, 8)$$

$$\text{Required area} = \int_{-1}^2 (3x + 2 - x^3) dx$$

$$= \frac{3}{2}(4 - 1) + 2(2 + 1) - \frac{1}{4}(16 - 1)$$

$$= \frac{9}{2} + 6 - \frac{15}{4} = \frac{18 + 24 - 15}{4} = \frac{27}{4}$$

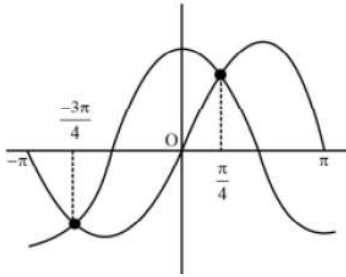
173. The area of the region enclosed by the curve $f(x) = \max\{\sin x, \cos x\}$, $-\pi \leq x \leq \pi$ and the x-axis is :

वक्र $f(x) = \max\{\sin x, \cos x\}$, $-\pi \leq x \leq \pi$ तथा x-अक्ष से घिरे क्षेत्र का क्षेत्रफल है :

- (A) 4 (B) $2(\sqrt{2} + 1)$ (C) $2\sqrt{2}(\sqrt{2} + 1)$ (D) $4(\sqrt{2})$

Ans. A

Sol.



Area =

$$\left| \int_{-\pi}^{-\frac{3\pi}{4}} \sin x dx \right| + \left| \int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}} \cos x dx \right| + \int_{\frac{\pi}{4}}^{\pi} \cos x dx + \int_{\frac{\pi}{4}}^{\pi} \sin x dx = 4$$

174. One of the points of intersection of the curves $y = 1 + 3x - 2x^2$ and $y = \frac{1}{x}$ is $\left(\frac{1}{2}, 2\right)$. Let the area of the region enclosed by the curves be $\frac{1}{24}(\ell\sqrt{5} + m) - n \log_e(1 + \sqrt{5})$, where $\ell, m, n \in \mathbb{N}$. Then $\ell + m + n$ is equal to :

वक्रों $y = 1 + 3x - 2x^2$ तथा $y = \frac{1}{x}$ का एक प्रतिच्छेदन $\left(\frac{1}{2}, 2\right)$ है। माना इन वक्रों से घिरे का क्षेत्रफल

$\frac{1}{24}(\ell\sqrt{5} + m) - n \log_e(1 + \sqrt{5})$ है, जहाँ $\ell, m, n \in \mathbb{N}$ है, तो $\ell + m + n$ का मान होगा :

- (A) 32 (B) 29 (C) 31 (D) 30

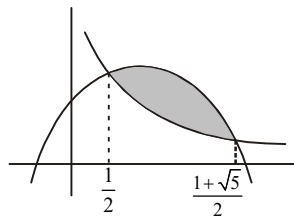
Ans. D

Sol. $2x^3 - 3x^2 - x + 1 = 0$
 $\Rightarrow (2x - 1)(x^2 - x - 1) = 0$

$$\Rightarrow x = \frac{1}{2}, x = \frac{1 + \sqrt{5}}{2}$$

$$\text{Area} = \int_{\frac{1}{2}}^{\frac{1 + \sqrt{5}}{2}} \left(1 + 3x - 2x^2 - \frac{1}{x} \right) dx$$

$$= \left[x + \frac{3x^2}{2} - \frac{2x^3}{3} - \ell \ln x \right]$$



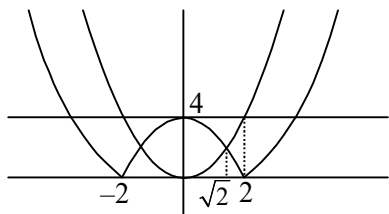
$$\begin{aligned}
 &= \frac{\sqrt{5}+1}{2} + \frac{3}{8}(\sqrt{5}+1)^2 - \frac{1}{12}(\sqrt{5}+1)^3 - \ln\left(\frac{\sqrt{5}+1}{2}\right) - \left(\frac{1}{2} + \frac{3}{8} - \frac{1}{12} - \ln\frac{1}{2}\right) \\
 &= \frac{1}{24} \left[12(\sqrt{5}+1) + 9(\sqrt{5}+1)^2 - 2(\sqrt{5}+1)^3 - 12 - 9 + 2 \right] - \ln\left(\frac{\sqrt{5}+1}{2} \times 2\right) \\
 &= \frac{1}{24} \left[12(\sqrt{5}+1) + 9(6+2\sqrt{5}) - 2(5\sqrt{5}+1+3\sqrt{5}(\sqrt{5}+1)-19) \right] - \ln(\sqrt{5}+1) \\
 &= \frac{1}{24} [14\sqrt{5}+15] - \ln(\sqrt{5}+1) \\
 \therefore l &= 14, m = 15, n = 1 \\
 l + m + n &= 30
 \end{aligned}$$

175. If the area of the region $\{(x, y) : |4 - x^2| \leq y \leq x^2, y \leq 4, x \geq 0\}$ is $\left(\frac{80\sqrt{2}}{\alpha} - \beta\right)$, $\alpha, \beta \in \mathbf{N}$, then $\alpha + \beta$ is equal to _____.

यदि क्षेत्र $\{(x, y) : |4 - x^2| \leq y \leq x^2, y \leq 4, x \geq 0\}$ का क्षेत्रफल $\left(\frac{80\sqrt{2}}{\alpha} - \beta\right)$, $\alpha, \beta \in \mathbf{N}$ है, तो $\alpha + \beta$ बराबर है _____.

Ans. 22

Sol.



$$A = \int_0^4 \sqrt{4+y} \, dy - \int_0^2 \sqrt{4-y} \, dy - \int_2^4 \sqrt{y} \, dy$$

$$= \left(\frac{(4+y)^{\frac{3}{2}}}{\frac{3}{2}} \right)_0^4 + \left(\frac{(4-y)^{\frac{3}{2}}}{\frac{3}{2}} \right)_0^2 - \left(\frac{y^{\frac{3}{2}}}{\frac{3}{2}} \right)_2^4$$

$$\frac{80\sqrt{2}}{3} - 16 = \frac{40\sqrt{2}}{3} - 16$$

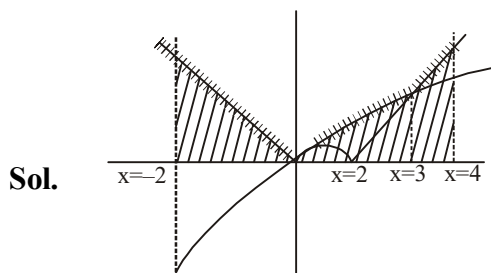
$$\alpha = 6, \beta = 16$$

$$\alpha + \beta = 22$$

176. The area of the region bounded by the curve $y = \max\{|x|, x|x-2|\}$, the x-axis and the lines $x = -2$ and $x = 4$ is equal to _____.

वक्र $y = \max\{|x|, x|x-2|\}$, x-अक्ष और रेखाओं $x = -2$, $x = 4$ से परिबद्ध क्षेत्र का क्षेत्रफल है _____.

Ans. 12

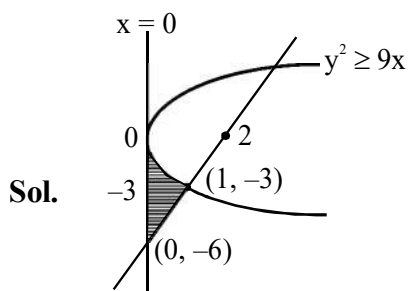


$$\begin{aligned} \text{Req. Area} &= \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 3 \times 3 + \frac{1}{2} \times 1 \times 1 \\ &= 12 \end{aligned}$$

177. Let the area of the bounded region $\{(x, y) : 0 \leq 9x \leq y^2, y \geq 3x - 6\}$ be A. Then 6A is equal to _____.

माना परिबद्ध क्षेत्र $\{(x, y) : 0 \leq 9x \leq y^2, y \geq 3x - 6\}$ का क्षेत्रफल A है। तो 6A बराबर है _____.

Ans. 15



$$A = \int_{-3}^0 \left(\frac{y^2}{9} - 0 \right) dy + \frac{1}{2} \cdot 1 \cdot 3$$

$$A = \left(\frac{y^3}{27} \right)_{-3}^0 + \frac{3}{2} = \frac{5}{2}$$

$$6A = 15$$

178. Let the function,

$$f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ a^2 + bx, & x \geq 1 \end{cases}$$

be differentiable for all $x \in \mathbb{R}$, where $a > 1$, $b \in \mathbb{R}$. If the area of the region enclosed by $y = f(x)$ and the line $y = -20$ is $\alpha + \beta\sqrt{3}$, $\alpha, \beta \in \mathbb{Z}$, then the value of $\alpha + \beta$ is _____.

माना कि फलन,

$$f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ a^2 + bx, & x \geq 1 \end{cases}$$

सभी $x \in \mathbb{R}$ के लिए अवकलनीय है, जहाँ $a > 1$, $b \in \mathbb{R}$ है। यदि $y = f(x)$ और रेखा $y = -20$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल $\alpha + \beta\sqrt{3}$, $\alpha, \beta \in \mathbb{Z}$ है, तो $\alpha + \beta$ का मान है _____.

Ans. 34

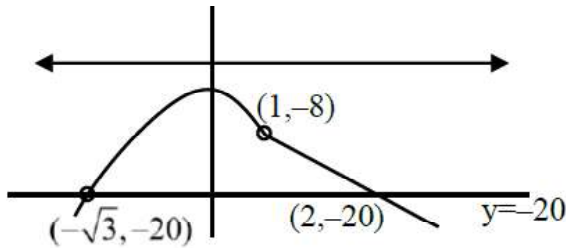
Sol. $f(x)$ is continuous and differentiable

at $x = 1$; LHL = RHL, LHD = RHD

$$-3a - 2 = a^2 + b, -6a = b$$

$$a = 2, b = -12$$

$$f(x) = \begin{cases} -6x^2 - 2; & x < 1 \\ 4 - 12x; & x \geq 1 \end{cases}$$



$$\text{Area} = \int_{-\sqrt{3}}^1 (-6x^2 - 2 + 20) dx + \int_1^2 (4 - 12x + 20) dx$$

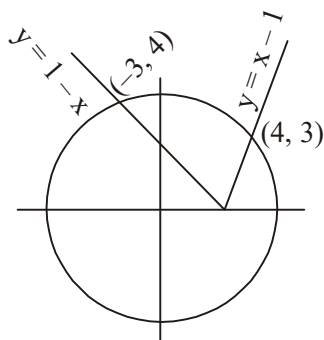
$$16 + 12\sqrt{3} + 6 = 22 + 12\sqrt{3}$$

179. If the area of the larger portion bounded between the curves $x^2 + y^2 = 25$ and $y = |x - 1|$ is $\frac{1}{4}(b\pi + c)$, $b, c \in \mathbb{N}$, then $b + c$ is equal to _____.

यदि वक्रों $x^2 + y^2 = 25$ और $y = |x - 1|$ के बीच परिबद्ध वृहत्तर भाग का क्षेत्रफल $\frac{1}{4}(b\pi + c)$, $b, c \in \mathbb{N}$ है, तो $b + c$ बराबर है

Ans. 77

Sol.



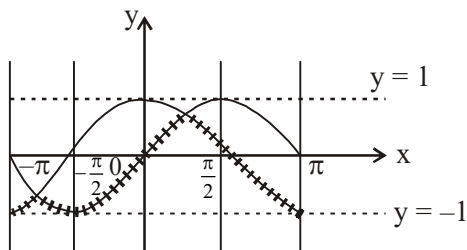
$$\begin{aligned}
 A &= 25\pi - \int_{-3}^4 \left(\sqrt{25 - x^2} - |x - 1| \right) dx \\
 &= 25\pi - \left[\frac{1}{2} x \sqrt{25 - x^2} + \frac{25}{2} \sin^{-1} \frac{x}{5} \right]_{-3}^4 + 8 + \frac{9}{2} \\
 &= \frac{1}{4} (75\pi + 2) \\
 \alpha &= 75 \quad \beta = 2 \\
 \alpha + \beta &= 77
 \end{aligned}$$

180. Let the area of the region enclosed by the curve $y = \min \{ \sin x, \cos x \}$ and the x-axis between $x = -\pi$ to $x = \pi$ be A. Then A^2 is equal to _____.

वक्र $y = \min \{ \sin x, \cos x \}$ तथा x-अक्ष के बीच परिवद्ध क्षेत्र का क्षेत्रफल A है। जहाँ x-अक्ष $x = -\pi$ तथा $x = \pi$ के बीच है, तो A^2 का मान होगा _____

Ans. 16

Sol. $y = f(x) = \min \{ \sin x, \cos x \}$



$$A = -\int_{-\pi}^{-\frac{3\pi}{4}} \cos x dx - \int_{-\frac{3\pi}{4}}^0 \sin x dx + \int_0^{\frac{\pi}{4}} \sin x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x dx$$

$$-\int_{\frac{\pi}{2}}^{\pi} \cos x dx$$

$$A = 4$$

$$A^2 = 16$$